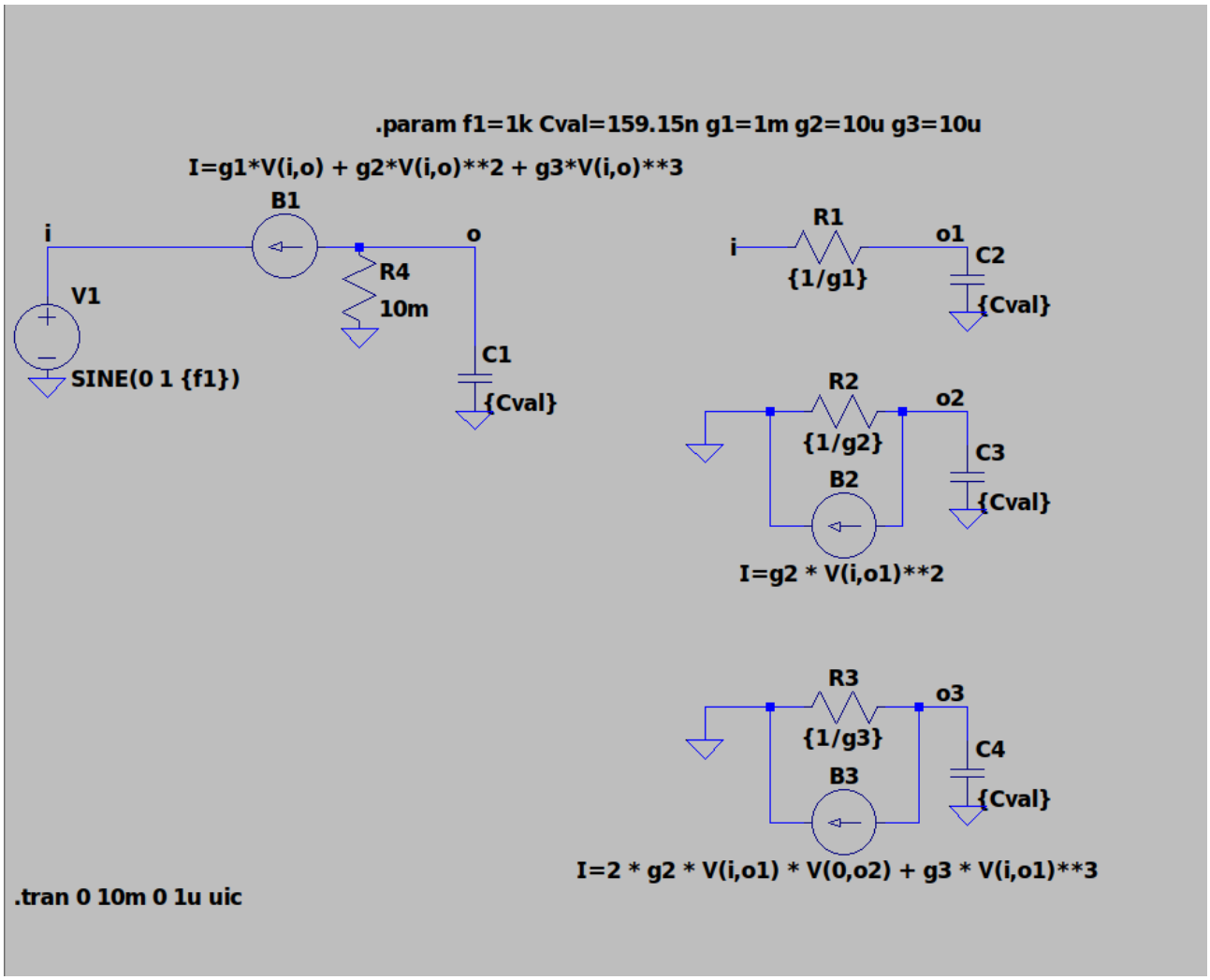
Week 11 Assignment

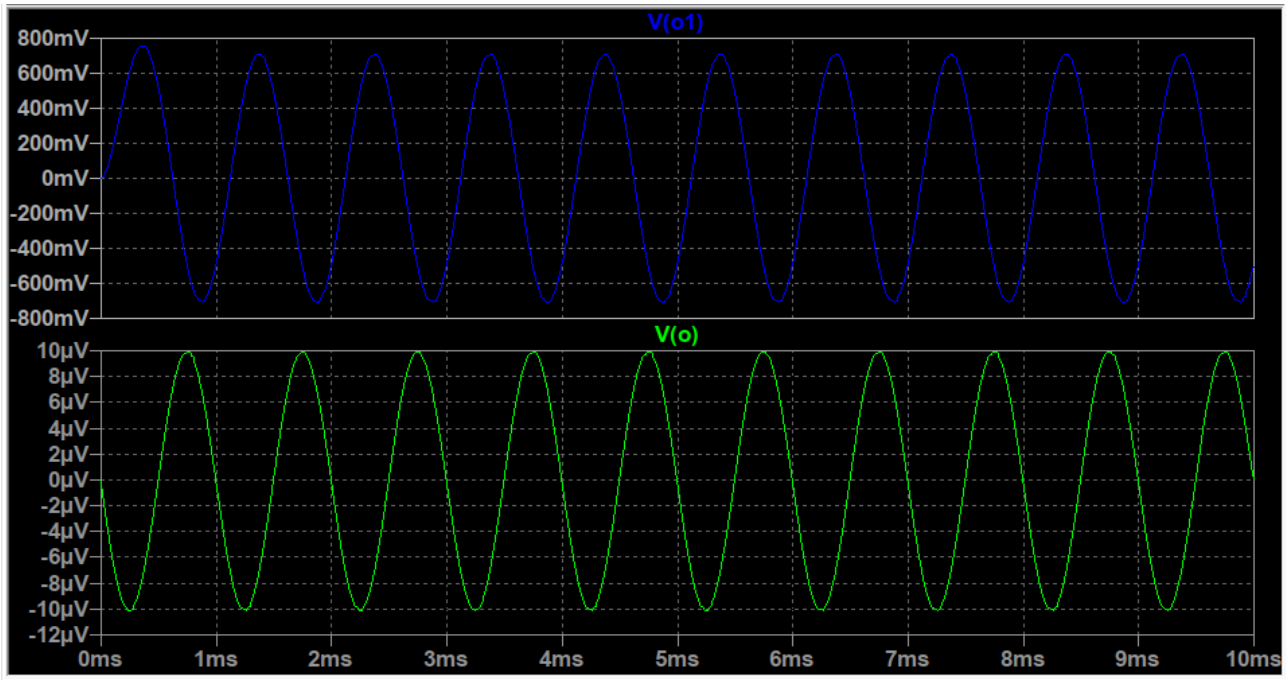
Problem 2

Created circuit



for case (a): $g_3 \approx 0$ and $g_2 = 10^{-5}$

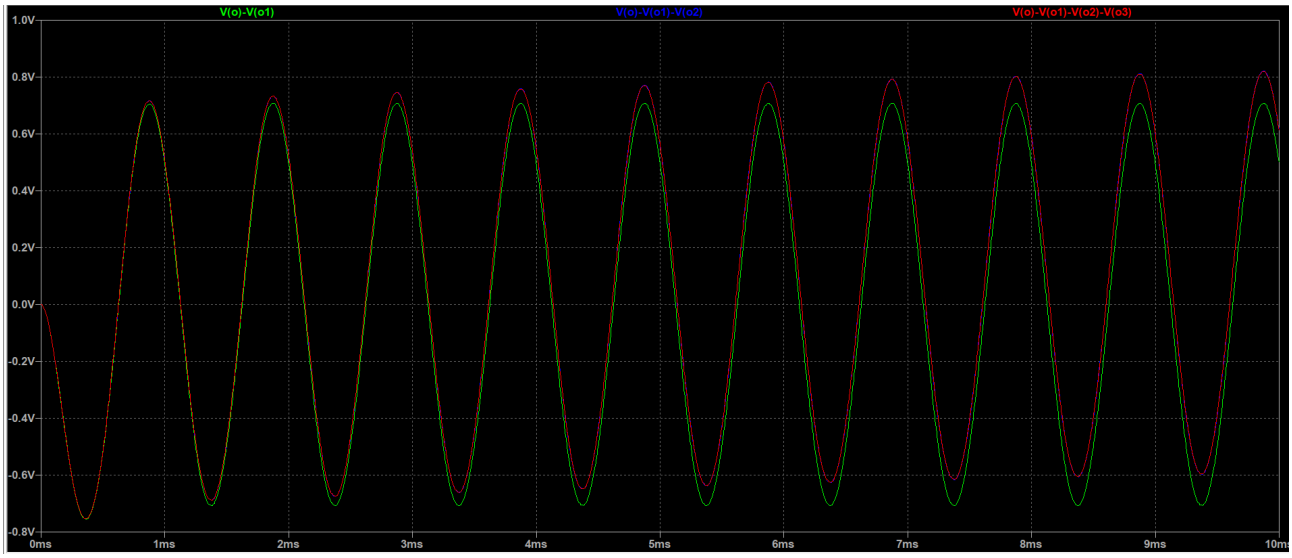
Plot of V_o and $V_{o(1)}$



we can see some phase reversal between them

Plot of

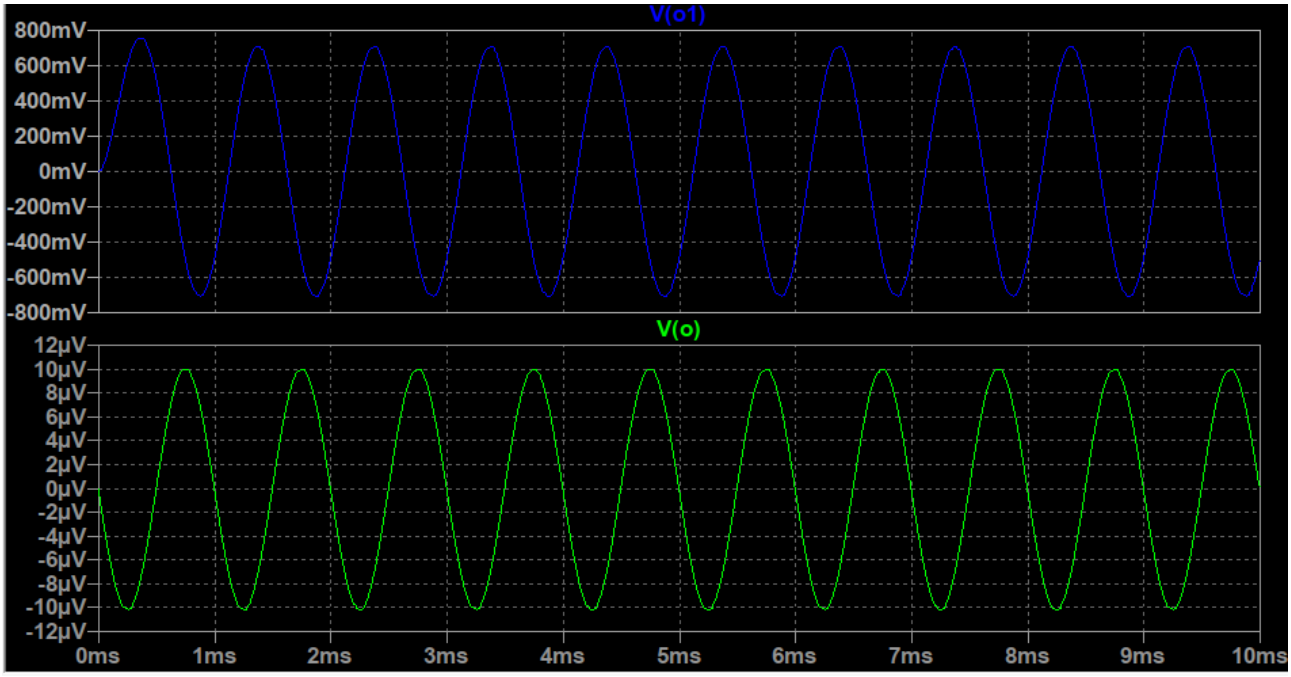
- $v_{o(t)} - v_{o(1)(t)}$
- $v_{o(t)} - v_{o(1)(t)} - v_{o(2)(t)}$
- $v_{o(t)} - v_{o(1)(t)} - v_{o(2)(t)} - v_{o(3)}(t)$



we can see effect of all terms here

for case (b): $g_2 = 10^{-5}$ and $g_3 = 10^{-5}$

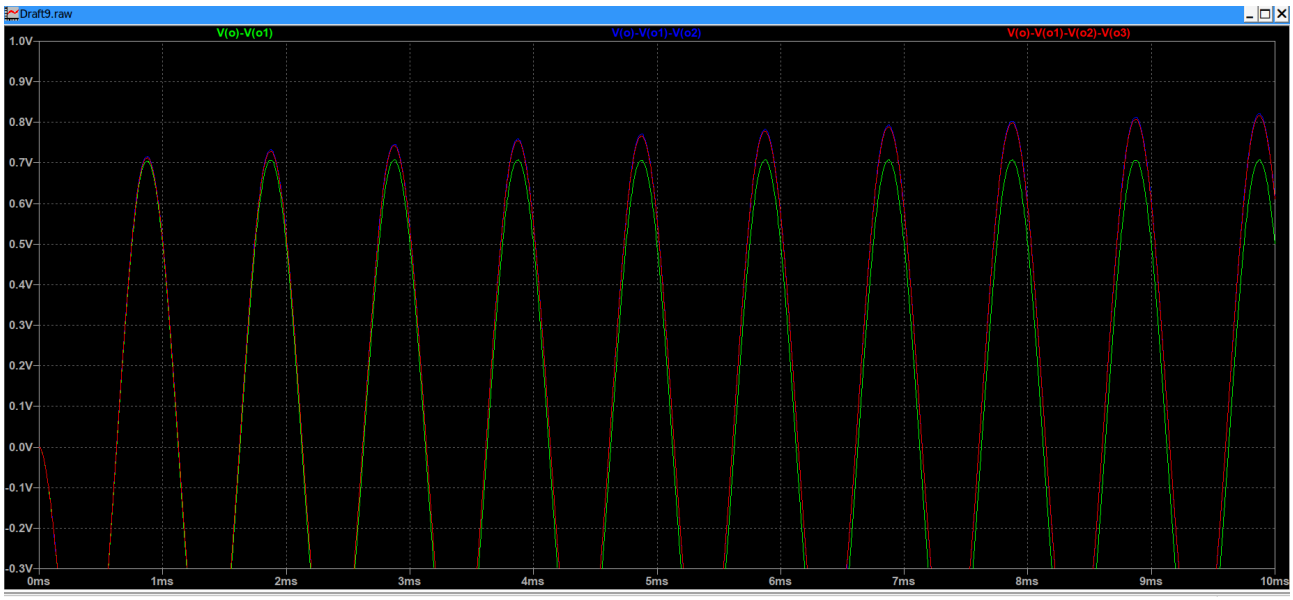
Plot of V_o and $V_{o(1)}$



we can see some phase reversal between them

Plot of

- $v_{o(t)} - v_{o(1)(t)}$
- $v_{o(t)} - v_{o(1)(t)} - v_{o(2)(t)}$
- $v_{o(t)} - v_{o(1)(t)} - v_{o(2)(t)} - v_{o(3)}(t)$

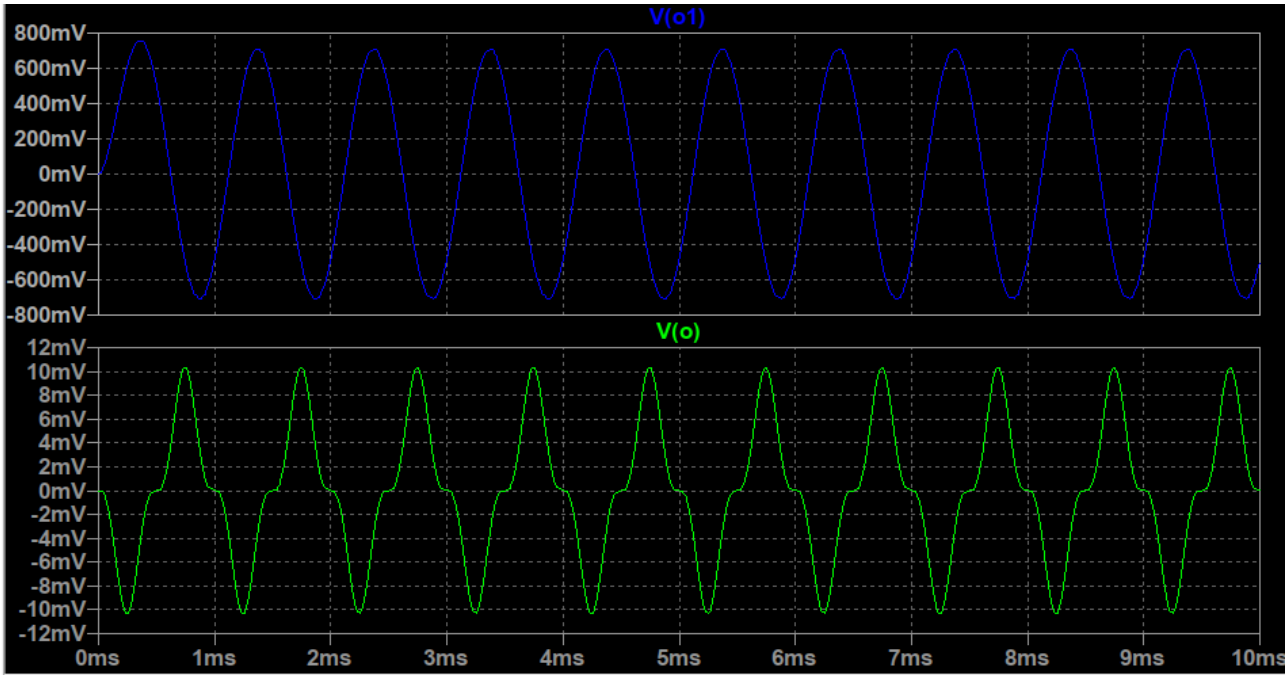


we can see effect of all terms

From this we can see that effect of non-linarity decreases with increase in order, with third order having minimum effect

for case (c): $g_2 \approx 0$ and $g_3 = 10^{-4}$

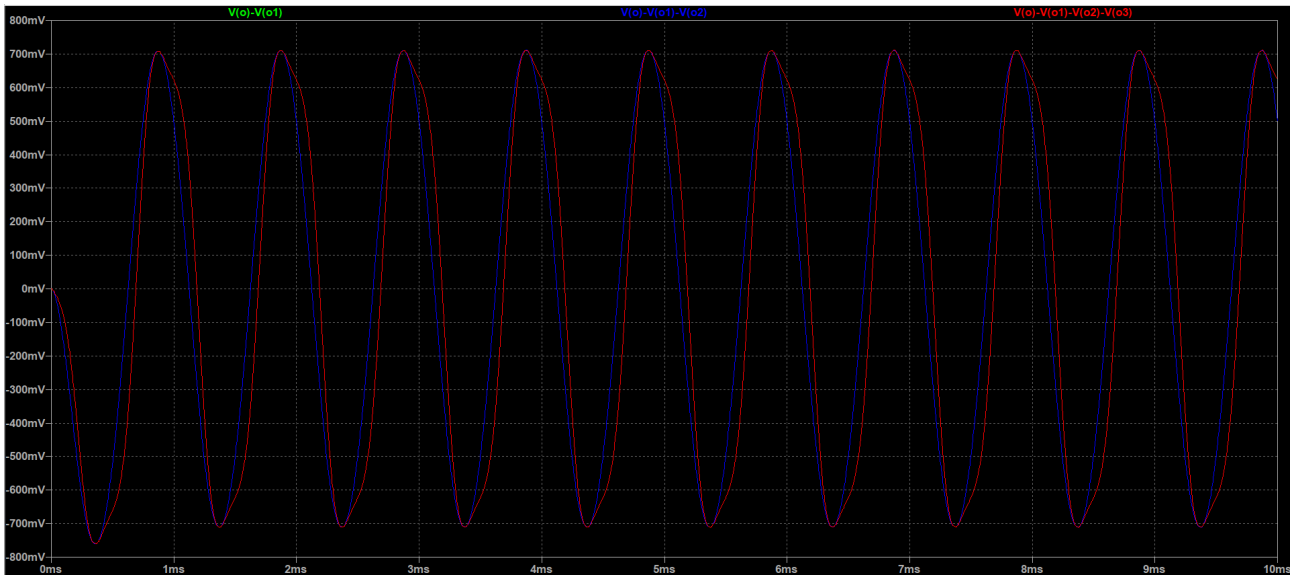
Plot of V_o and $V_{o(1)}$



we see some phase reversal with distortion in final output

Plot of

- $v_{o(t)} - v_{o(1)(t)}$
- $v_{o(t)} - v_{o(1)(t)} - v_{o(2)(t)}$
- $v_{o(t)} - v_{o(1)(t)} - v_{o(2)(t)} - v_{o(3)}(t)$



in this we see 3rd order has highest visible effect on output compared as 2nd order effect is zero